ANDROID APP DEVELOPMENT · TASK 1

Responsive Personal Profile App

Android fundamentals, Kotlin basics, XML layouts, Activities & Intents, resources and Material Design 3

7	16	10	M3
screens	tests passing	Kotlin files	Material Design

DEVELOPER
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PROGRAMME
Sumerix Global · Android App Development · 36 weeks

BATCH START
15 October 2026

STACK
Kotlin 2.0 · AGP 8.7 · Material 3 · minSdk 26 / target 35

1 · Overview

Android fundamentals, Kotlin basics, XML layouts, Activities & Intents, resources and Material Design 3. Package `com.eswar.profile`.

- Splash screen → Profile → About / Skills / Contact, all separate Activities connected with explicit Intents (title + source passed as extras)
- Material 3 theme with light and dark colour sets (`values-night`), cards, chips, progress bars and buttons
- Responsive: `layout-sw600dp` two-pane tablet profile and `values-sw600dp` dimensions
- Reusable layouts with `` (hero, stats row, section cards) and item layouts for skills / timeline
- Skills with progress bars, experience & education timeline, share-profile and contact (mail / LinkedIn / GitHub) implicit Intents
- All text, colours, dimensions and styles in resources — nothing hard-coded

7
screens captured

16
automated tests · 0 failing

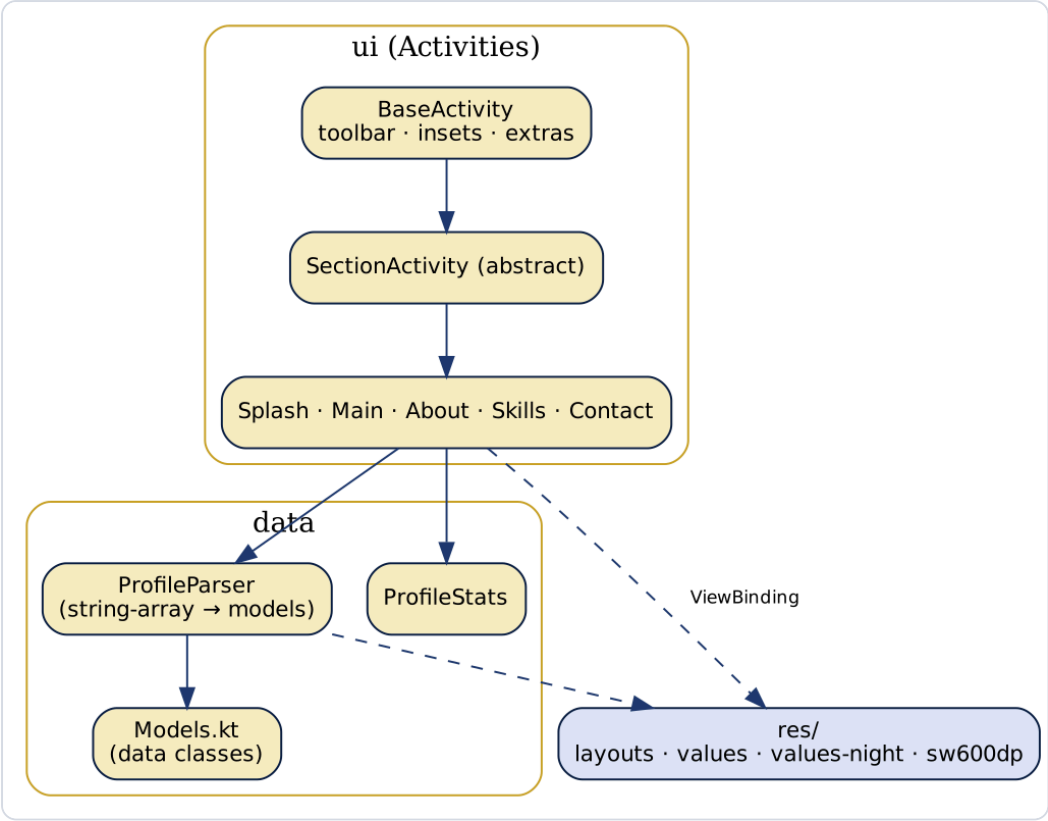
10
Kotlin source files

15
XML layouts

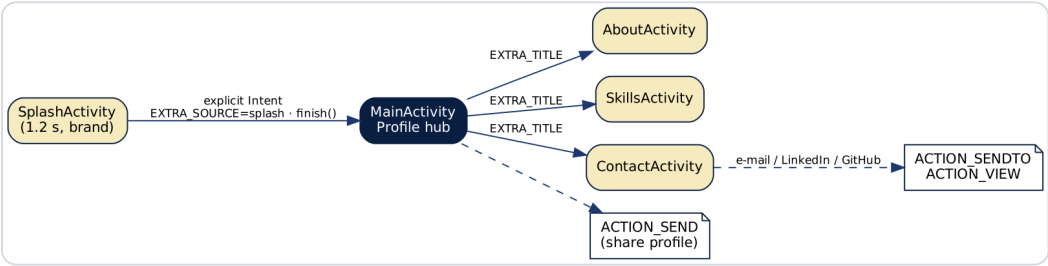
2 · Brief coverage

Brief part	Implementation
A – Android development	Architecture diagram below; minSdk 26 / target 35, AndroidX + Jetpack
B – Android Studio & Gradle	Gradle Kotlin DSL, version catalogue-style plugin block, wrapper 8.14.3
C – Kotlin fundamentals	data/Models.kt (data classes, null safety), ProfileParser.kt (functions, loops, when), ProfileStats.kt
D – Project structure	Manifest, java/, res/ (drawable, layout, values, values-night, mipmap) — tree below
E – XML layouts	ConstraintLayout, LinearLayout, FrameLayout, ScrollView (+ MaterialCardView, TextInputLayout)
F – Views	TextView, Button, ImageView, CardView/MaterialCardView, chips, progress indicators
G – Activities	5 Activities, abstract SectionActivity base, Intent extras, lifecycle-aware BaseActivity
H – Resources	strings, colors (day/night), themes, dimens (phone/tablet), vector drawables, arrays
I – Material Design	Material 3 DayNight theme, MaterialToolbar, cards, buttons, chips
J/K – Mini project & industry challenge	Splash, Profile, About, Skills, Contact, navigation, Material UI, responsive layout

3 · Architecture

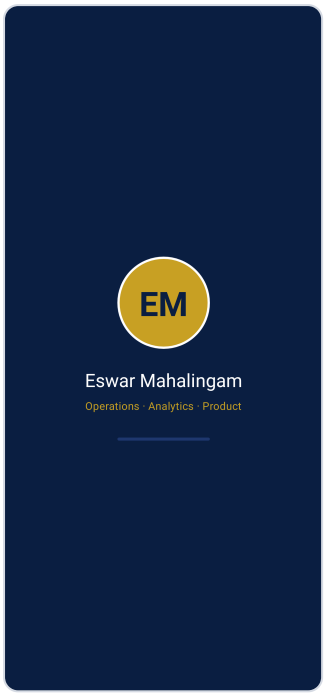


Architecture

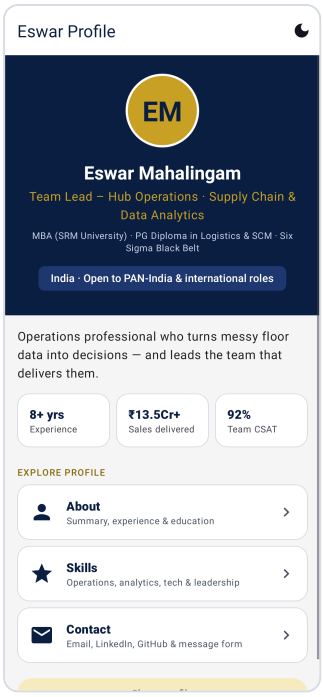


Intent navigation flow

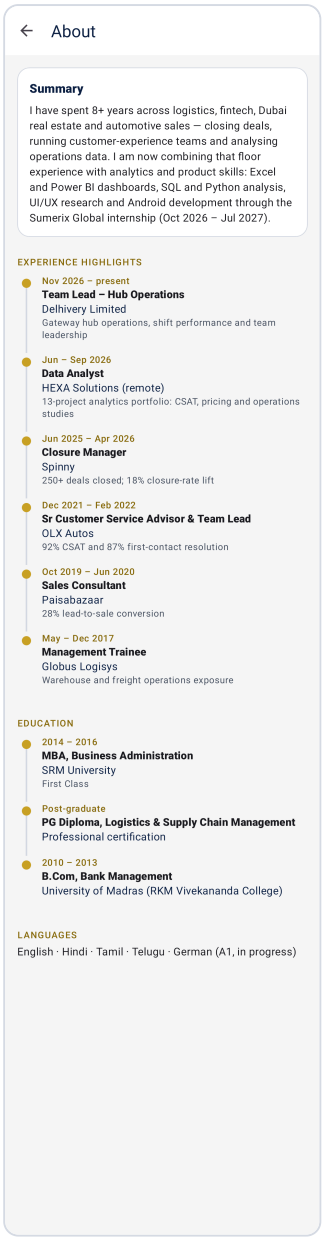
4 · Screens (phone, light & dark)



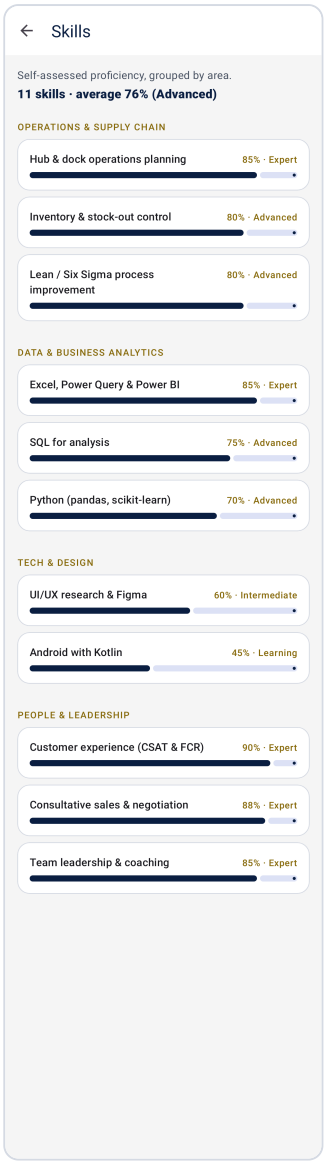
Splash



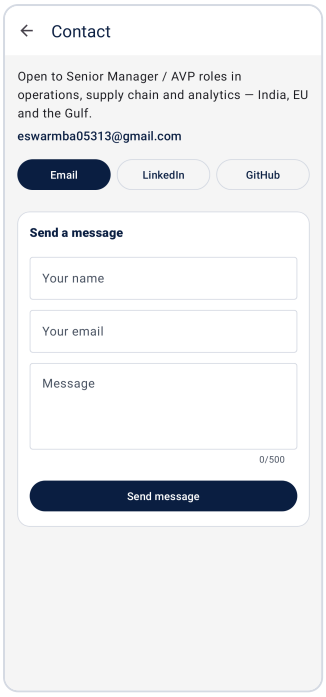
Profile



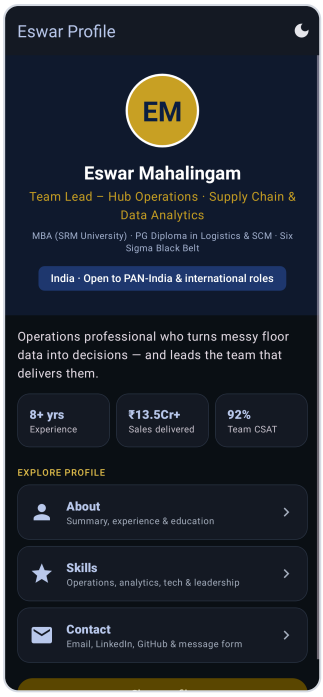
About



Skills

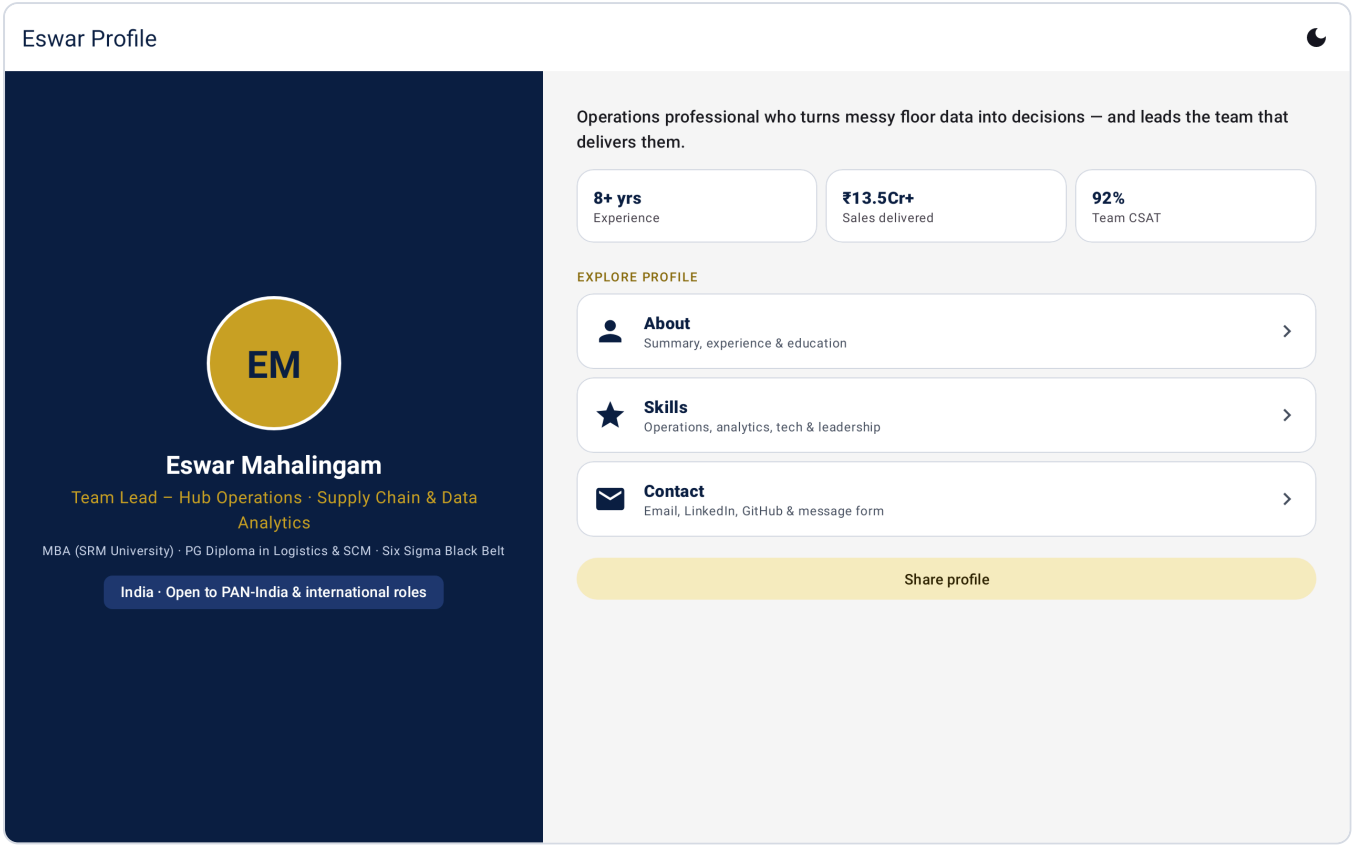


Contact



Profile dark

5 · Tablet layout (sw600dp)



Profile tablet

6 · Testing

Test class	Tests	Failures	Cases
ProfileLogicTest	5	0	groupsByCategoryInEnumOrder, parsesSkillsAndSkipsMalformedRows, averageAndLabels, parsesTimelineWithOptionalDetail, initialsAndEmailValidation
NavigationTest	4	0	contactFormValidatesBeforeSending, splashOpensProfileAfterDelay, sectionShowsTitlePassedThroughIntent, profileCardsOpenSectionsWithTitleExtra
ScreenshotTest	7	0	phone_04_skills, tablet_07_profile, phone_06_profile_dark, phone_03_about, phone_02_profile, phone_05_contact, phone_01_splash

Tests run on the JVM with Robolectric (API 34). Screenshot tests render every screen with Roborazzi in native graphics mode, so the images in this document are produced by the real XML layouts, themes and resources.

7 · Project structure

```
src/
  main/
    AndroidManifest.xml
    java/
      com/
        eswar/
          profile/
            data/
              Models.kt
              ProfileParser.kt
              ProfileStats.kt
            ui/
              AboutActivity.kt
              BaseActivity.kt
              ContactActivity.kt
              MainActivity.kt
              SectionActivity.kt
              SkillsActivity.kt
              SplashActivity.kt
    res/
      drawable/
        bg_avatar.xml
        bg_timeline_dot.xml
        ic_chevron.xml
        ic_dark_mode.xml
        ic_launcher_foreground.xml
        ic_mail.xml
        ic_person.xml
        ic_star.xml
      layout/
        activity_about.xml
        activity_contact.xml
        activity_main.xml
        activity_skills.xml
        activity_splash.xml
        include_profile_hero.xml
        include_section_cards.xml
        include_stats_row.xml
        item_section_card_about.xml
        item_section_card_contact.xml
        item_section_card_skills.xml
        item_skill.xml
        item_skill_header.xml
        item_timeline.xml
      layout-sw600dp/
        activity_main.xml
      menu/
        menu_main.xml
      mipmap-anydpi-v26/
        ic_launcher.xml
      values/
        arrays.xml
        colors.xml
        dimens.xml
        ic_launcher_background.xml
        strings.xml
        themes.xml
      values-night/
        colors.xml
      values-sw600dp/
        dimens.xml
  test/
```

```
    java/
      com/
        eswar/
          profile/
            ProfileLogicTest.kt
            ui/
              NavigationTest.kt
              ScreenshotTest.kt
    resources/
```